GENERATIVE ARTIFICIAL INTELLIGENCE

A simple introduction to GenAI, how it works, and where it is used

1. What Is Generative AI?

Generative Artificial Intelligence (Generative AI or GenAI) is a type of AI that creates new content based on patterns learned from large collections of data. It can generate text, images, music, audio, video, computer code, and other forms of content. Unlike systems designed only to classify or predict, generative models produce outputs that resemble examples they learned from.

Examples: A chatbot can draft an email, an image model can create an illustration from a text prompt, and a coding assistant can suggest a Python function. These tools can help people work faster, but their output should still be reviewed.

2. How Does Generative AI Work?

GenAI models are trained using many examples. During training, a model learns relationships and patterns in the data. A language model, for example, learns how words and ideas commonly relate. When a user sends a prompt, the model uses its learned patterns to generate a response. The result is generated—not simply copied from a conventional database.

A typical workflow:

- **Input:** The user provides a prompt, question, document, or other context.
- **Processing:** The model interprets the input and uses learned patterns.
- **Generation:** The model produces text, code, an image, or another output.
- **Review:** A person checks the result for accuracy, quality, and suitability.

3. Important GenAI Concepts

Large Language Model (LLM): A model trained to understand and generate language. LLMs are used for chatbots, summarization, translation, and coding assistance.

Prompt: The instruction or context given to a model. A good prompt can specify the task, audience, tone, format, and limits to get a more useful result.

Tokens: Small pieces of text processed by a language model. A token may be a word, part of a word, or punctuation. Model input and output limits are often measured in tokens.

Temperature: A setting that can affect variation. Lower values generally encourage more consistent responses, while higher values can make responses more varied.

GenAI is useful, but it can sometimes produce inaccurate or made-up details. Always verify important information, especially before using it for high-impact decisions.

GENAI IN PRACTICE

Applications, RAG, benefits, limitations, and responsible use

4. Common Applications of GenAI

- **Education:** Explaining concepts, creating practice questions, and supporting study.
- **Business and customer support:** Drafting replies, summarizing issues, and sorting requests.
- **Software development:** Suggesting code, explaining errors, generating tests, and documentation.
- **Content creation:** Drafting articles, presentations, product descriptions, and social posts.
- **Design and media:** Generating or editing images, audio, video, and creative concepts.
- **Research and productivity:** Summarizing documents, extracting key points, and brainstorming.

5. What Is Retrieval-Augmented Generation (RAG)?

Retrieval-Augmented Generation (RAG) connects a generative model to external information. Rather than relying only on its training, a RAG application searches a collection of documents or a database, retrieves relevant passages, and supplies them as context for the model to answer a question.

Basic RAG flow: Documents → text chunks → embeddings → vector database → retrieve relevant context → LLM generates an answer.

For example, a user can upload a company policy PDF and ask about leave rules. The system retrieves the relevant text and generates an answer grounded in that document. RAG can improve relevance and traceability, but it does not guarantee that every answer is correct.

6. Benefits and Limitations

Benefits

- Saves time on repetitive writing and information tasks.
- Helps people brainstorm and explore ideas.
- Makes natural-language interaction with software easier.
- Can adapt output to different formats and audiences.

Limitations

- May provide incorrect, outdated, or fabricated information.
- Can reflect bias present in training data.
- May misunderstand vague prompts or missing context.
- Sensitive information should not be shared unless the tool and its privacy settings are appropriate.

7. Responsible Use of GenAI

Use GenAI as an assistant—not as an unquestioned authority. Write clear prompts, check facts against reliable sources, protect private data, respect copyright, and keep human oversight for important decisions. The best results come when people combine AI's speed with human judgment and domain knowledge.

In summary: Generative AI is a technology for creating new content from learned patterns. Understanding prompts, models, RAG, and responsible use helps users build practical and trustworthy AI applications.